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CS 600

Q1. No. 19.8.3

Suppose a certain birth defect occurs independently at random with probability $p = 0.02$ in any live birth. Use a Chernoff bound to bound the probability that more than 4% of the 1 million children born in a given large city have this birth defect.

Ans.

For $i = 1, \dots, 10^6$, Compute μ

$$\mu = E[X] = \sum_{i=1}^{1000000} E[X_i] = \sum_{i=1}^{1000000} (0.02) = 20000$$

4% of 1 million children would be $= 0.04 * 1000000 = 40000$

By Chernoff bounds, for $\delta = 1$, upper bound is

$$Pr(X \geq (1+\delta)\mu) = P(X \geq 40000) < [e / (1 + \delta)]^{\delta\mu} = [e / 4]^{20000}$$

Q2. No. 19.8.18

- (a) Consider a modification of the Fisher-Yates random shuffling algorithm where we replace the call to $\text{random}(k+1)$ with $\text{random}(n)$, and take the for-loop down to 0, so that the algorithm now swaps each element with another element in the array, with each cell in the array having an equal likelihood of being the swap location. Show that this algorithm does not generate every permutation with equal probability.

Hint: Consider the case when $n = 3$.

Ans.

Let's analyze this with a small example:

Suppose the input array is [A, B, C], and we set $n=3$

We examine the behavior of the modified Fisher-Yates algorithm that:

- Iterates from $i=2$ to $i=0$,
- Swaps element at position i with a randomly chosen index $j \in \{0,1,2\}$ (i.e., $\text{random}(n)$ instead of $\text{random}(i+1)$).

Step-by-step analysis for all possible outcomes:

Let's examine all possible execution paths (with uniform randomness):

Step 1: $i = 2$, choose $j \in \{0, 1, 2\}$

There are 3 choices:

- Swap 2 with 0: [C, B, A]
- Swap 2 with 1: [A, C, B]
- Swap 2 with 2: [A, B, C]

Step 2: $i = 1$, choose $j \in \{0, 1, 2\}$

For each result from Step 1, again swap index 1 with any index in 0-2. Total of $3 \times 3 = 9$ possible paths.

But not all resulting permutations are equally likely. For example:

- [A, B, C] appears in more than one path.
- [B, A, C] may appear in fewer paths.
- [C, B, A] might only be generated in 1 way.

You can count and verify that some permutations occur multiple times in this tree of swaps, while others occur only once.

The modified algorithm does not generate every permutation with equal probability. It breaks the uniform distribution guaranteed by the original Fisher-Yates shuffle. This is evident even in the small case of $n=3$, where not all 6 permutations are generated with equal likelihood.

Q3. No. 19.8.35

In a famous experiment, Stanley Milgram told a group of people in Kansas and Nebraska to each send a postcard to a lawyer in Boston, but they had to do it by forwarding it to someone that they knew, who had to forward it to someone that they knew, and so on. Most of the postcards that were successfully forwarded made it in 6 hops, which gave rise to the saying that everyone in America is separated by "six degrees of separation." The idea behind this experiment is also behind a technique, called **probabilistic packet marking**, for doing traceback during a distributed denial-of-service attack, where a website is bombarded by connection requests. In implementing the probabilistic packet marking strategy, a router, R , will, with some probability, $p \leq 1/2$, replace some seldom-used parts of a packet it is processing with the IP address for R , to enable tracing back the attack to the sender. It is as if, in the Milgram experiment, there is just one sender, who is mailing multiple postcards, and each person forwarding a postcard would, with probability, p , erase the return address and replace it with his own. Suppose that an attacker is sending a large number of packets in a denial-of-service attack to some recipient, and every one of the d routers in the path from the sender to the recipient is performing probabilistic packet marking.

- (a) What is the probability that the router farthest from the recipient will mark a packet and this mark will survive all the way to the recipient?
- (b) Derive a good upper bound on the expected number of packets that the recipient needs to collect to identify all the routers along the path from the sender to the recipient.

Ans.

(a) Probability that the farthest router marks a packet and the mark survives to the recipient

In the probabilistic packet marking strategy, each router marks a packet with probability $p \leq 1/2$. Suppose there are d routers on the path from sender to recipient, and router i (where $1 \leq i \leq d$, with $i=1$ being the sender and $i=d$ the last router before the recipient) marks the packet.

Let us consider the farthest router from the recipient i.e, the first router (router 1).

The probability that this router marks the packet is p . Then, for the mark to survive through the remaining $d-1$ routers (who might overwrite it), each of them must not mark the packet. That happens with probability $(1-p)$ per router.

Thus, the probability that router 1's mark survives to the recipient is:

$$P = p \cdot (1-p)^{d-1}$$

In general, for the i -th router (counted from sender to recipient), the probability that its mark is received by the recipient is:

$$P_i = p \cdot (1-p)^{d-i}$$

(b) Upper bound on the expected number of packets needed to identify all routers (Coupon Collector Analogy)

This problem is analogous to the coupon collector problem, where the recipient tries to collect markings from all d distinct routers.

Let X be the random variable representing the number of packets needed to receive identifiable markings from all d routers.

We model this as:

$$X = X_1 + X_2 + \dots + X_d$$

Where X_i is the number of packets needed to discover a new router given that $i-1$ distinct routers are already known.

At any stage, the probability of receiving a new router's mark is:

$$P_i = \frac{d - (i-1)}{d} = \frac{d-i+1}{d}$$

Thus, the expected number of packets to get the i -th new router is:

$$E[X_i] = 1 / P_i = d / (d-i+1)$$

Using linearity of expectation, the total expected number of packets is:

$$E[X] = \sum_{i=1}^d \frac{d}{(d-i+1)} = d \cdot \sum_{k=1}^d \frac{1}{k} = d \cdot H_d$$

Where H_d is the d -th harmonic number, and we know:

$$\ln d < H_d < \ln d + 1$$

Therefore, the expected number of packets is:

$$E[X] \leq d \cdot (\ln d + 1)$$